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I Have No Mouth, and I Must Rhyme: Uncovering Internal Phonetic Representations in LLaMA 3.2

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Abstract

Large language models demonstrate proficiency on phonetic tasks, such as rhyming, without explicit phonetic or auditory grounding. In this work, we investigate how `Llama-3.2-1B-Instruct` represents token-level phonetic information. Our results suggest that Llama uses a rich internal model of phonemes to complete phonetic tasks. We provide evidence for high-level organization of phoneme representations in its latent space. In doing so, we also identify a "phoneme mover head" which promotes phonetic information during rhyming tasks. We visualize the output space of this head and find that, while notable differences exist, Llama learns a model of vowels similar to the standard IPA vowel chart for humans, despite receiving no direct supervision to do so.

1. Introduction

Language models ("LMs") cannot hear speech. In spite of this, many large LMs can consistently produce rhymes, poetry, alliteration, and other phenomena which seem to require a fundamental understanding of phonetic properties. We propose that these models complete these tasks using underlying token-level phonetic representations as well as circuits to retrieve phonetic information.

Up to this point, study of LMs' phonetic behavior has been largely limited to training models on grounded phonetic information (English et al., 2023; Popescu-Belis et al., 2023). Anthropic's recent work *On the Biology of a Large Language Model* (Lindsey et al., 2025) uses attribution graphs to provide evidence of rhyme planning circuits in Claude 3.5 Haiku, but they do not rigorously study the internal phonetic representations that allow these circuits to function.

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It is reasonable to ask the extent to which an LM is able to infer phonetic properties directly from tokens. Following results that LMs and even simple text models can extract information about the world without direct supervision (Louwerse and Benesh, 2012; Mikolov et al., 2013; Gurnee and Tegmark), we hypothesize that LMs also learn a robust, structured model of phonetic information that supports interventions. We investigate how Llama 3.2 (Grattafiori et al., 2024) represents phonetic information through methods commonly used in interpretability analysis and find evidence to support this hypothesis: rather than simply memorizing phonetic information for various tokens, Llama uses structure in its latent space across tokens to represent the phonetics of given input tokens.

Our experiments follow a straightforward methodology. To explore linear token-level phonetic information, we employ linear probes to identify subspaces of the residual stream and embedding and inspect the subspaces. To explore the mechanisms that leverage this information, we isolate components of the model which prove to be impactful in expressing the LM's phonetic beliefs. Using this simple approach, we find evidence of rich phonetic representations within `Llama-3.2-1B-Instruct`.

In Section 2, we find vectors in the embedding space corresponding to common English phonemes. We perform causal interventions in the embedding space using these vectors to alter the model's performance on a rhyming task, demonstrating their role in rhyming processes. In Section 3 we identify a single "phoneme mover head" using activation patching. We decode this head's result vectors using logit lens (nostalgebraist, 2020) and demonstrate that the phonetic information in this head is, to some extent, cross-lingual. In Section 4, we use the result vectors of this phoneme mover head to perform a phonetically-informed dimensionality reduction on the embedding space in order to analyze the geometry of the phoneme vectors from Section 2. We see consistent linear patterns across internal representations of vowel phonemes (Figure 1).

These patterns differ from anatomical properties of vowel phonemes, suggesting the presence of a robust internal vowel model which is distinct from human vowel models. We construct an organized representation of these patterns

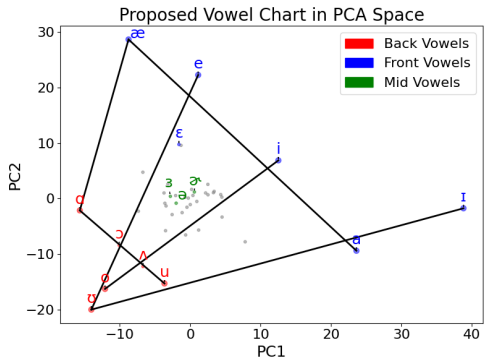

Figure 1: Patterns emerge from vowel representations under our methodology, revealing a world model partially inconsistent with human anatomy.

and contrast it with existing anatomically-grounded vowel organization systems.

2. Phonetic Information in Token Embeddings

In Llama 3.2, like in most open-source LMs, the tokenization process does not explicitly encode any phonetic information about input tokens. We hypothesize that some phonetic information is encoded through token embeddings. To investigate this hypothesis, we train a multi-hot linear probe to predict which phonemes are present in a word from its embedding. We differentiate phonemes based on the International Phonetic Alphabet, using WikiPron (Lee et al., 2020) as a pronunciation reference.

Probing the embedding space Our probe predicts the correct phonemes for approximately 96% of single-token words, compared with 42% for the same probe architecture trained on embeddings from a randomly-generated embedding matrix. This indicates that the model's embedding matrix encodes some amount of recoverable phonetic information.

Since this probe is linear, it essentially constitutes a linear map from Llama 3.2's 2048-dimensional embedding space to what we call an "IPA phoneme space": a 44-dimensional space with one axis for each common English-language phoneme. Each row of our probe matrix, therefore, could be considered a representation of its corresponding phoneme in the embedding space.

Causal interventions on embeddings To test the degree to which the rows of our probe represent phonemes in latent space, we run a causal intervention experiment to change the expected rhyming output for a given target word by intervening on embedding vectors. We test the model on a simple rhyming task:

```
prompt = """Here are a few examples of words
that rhyme with <word>:"""
```

where `<word>` is a single-token word with one unique vowel sound. We then select two "phoneme vectors" (rows of our probe matrix) in the embedding space: ξ, the phoneme vector corresponding to the vowel in `<word>`; and μ, the phoneme vector corresponding to a different vowel. We perform a forward pass of Llama with the following intervention at the embedding step:

$$E = E + c(\mu - \xi)$$

where E is the embedding vector corresponding to our rhyming word `<word>`, and c is a scalar. As we increase c (the weight of our intervention), the model's prediction tends to switch from words with the ξ vowel to words with the μ vowel. Figure 2 shows example model output results where `<word>` is `leet` pronounced /liːt/, ξ corresponds to /i/, and μ corresponds to ɛ.

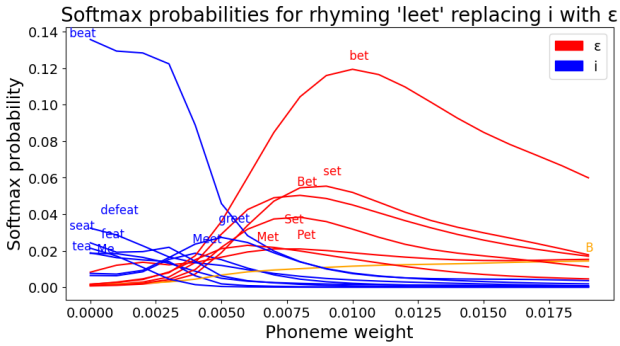

Figure 2: Example intervention on rhymes with `leet`. Blue indicates that the ξ phoneme /i/ is present, red indicates that the μ phoneme /ɛ/ is present.

3. Phoneme Mover Head

BOS Here are a few examples of words that rhyme with grab : *
→ **Promoted by head**: Gab, Bob, ブ, bara, Barbara, bo, gado, boot, Bob, b
→ **Top predictions**: Cab, cab, dab, Tab, Crab, Grab, Br, tab, Sab, b
BOS Here are a few examples of words that rhyme with plush : *
→ **Promoted by head**: ush, uss, ブ, uch, us, usk, usses, usc, ش, USH
→ **Top predictions**: crush, lush, h, Bush, flush, splash, plush, rush, crunch, Crush
BOS Here are a few examples of words that rhyme with clean : *
→ **Promoted by head**: enic, lean, enal, enas, ean, Wein, ান, Ben, LEAN, Neil
→ **Top predictions**: keen, clean, Screen, green, Seen, serene, seen, screen, Green, Clean

Figure 3: Across three examples: Attention patterns of H13L12 at the final token shows the head attends to the rhyme target token. Running logit lens on the corresponding result vector (shown after →) produces phonetically similar tokens.

Given that embeddings seem to contain phonetic information, we seek to identify model components which use this

information to complete rhyming tasks. We run activation patching experiments over all attention heads and MLP components across all layers.

Patching setup To patch, we perform two full passes of the model using parallel prompts containing two sufficiently different[1] rhyme target words. For example, setting `<word>="clean"` in our above template results in the top predicted token `"keen"`. Similarly, setting `<word>="track"` results in `"back"`. These then form our "clean" and "corrupted" runs (in line with Meng et al. (2023)) allowing us to inspect the mean normalized logit difference between `"keen"` and `"back"`. Head 13 of Layer 12 ("H13L12") emerged as the most critical for our rhyming task with a mean normalized logit difference of 0.48 (out of 1). This was significantly higher than both the mean (0.002) and the second highest value (0.19). See Appendix B for a visual representation.

Result vectors and cross-lingual features To understand the effect of this head on task completion, we explore its contributions to the residual stream by inspecting its result vectors[2]. We decode these vectors into the vocab space using logit lens (nostalgebraist, 2020) in order to study the tokens this head promotes. Figure 3 demonstrates these results along with last-token attention patterns. We detail further study of the head dimension in Appendix C.

There is a clear, if approximate, phonetic similarity between the target rhyme word and the decoded result vectors of H13L12. These results and the associated attention patterns suggest that this head moves phonetic information from the target word to the final token's residual stream. Because of this apparent phonetic promotion behavior, we call this head a "phoneme mover head". We investigate the connection between the result vectors produced by the head and completion of our rhyming task in Appendix D.

We also see clear evidence of the LM modeling phonetic information across languages, as decoding result vectors produces language-agnostic sets of similar phonemes. In Figure 3, for instance, we see that one of the top tokens promoted into the residual stream for `plush` is "sh (Arabic)" /ʃ/, for `clean` we see "een (Hindi)" (/iːn/), and for `grab` we see "bu (Japanese)" (/bɯ/), among others.

Further Phonetic Heads Upon further inspection we discovered a set of heads (H13L12, H21L14, H22L14) with nearly identical attention patterns. After zero ablating all three of these heads, we noticed that the model could no longer correctly produce a *single* rhyming token. Instead, it produced a first token (typically a single letter) and then a corresponding second token which typically completed the response word (For example if the target word was "plush" we might get "l" "ush"). Results were generally mixed as to if the response produced was truly a correct rhyme or not. Omitting any one of these three heads from ablation reinstated the model's original rhyming ability. Of further interest is that the composition scores (Elhage et al., 2021) for all three of these heads and all prior heads were also essentially identical, suggesting a common channel.

[1]Two words are "*sufficiently different*" iff there is no third word that rhymes with both simultaneously.

[2]i.e. the projection of a single head back to the residual stream dimension through its output matrix, see Elhage et al. (2021)

4. Geometry of Phoneme Vectors

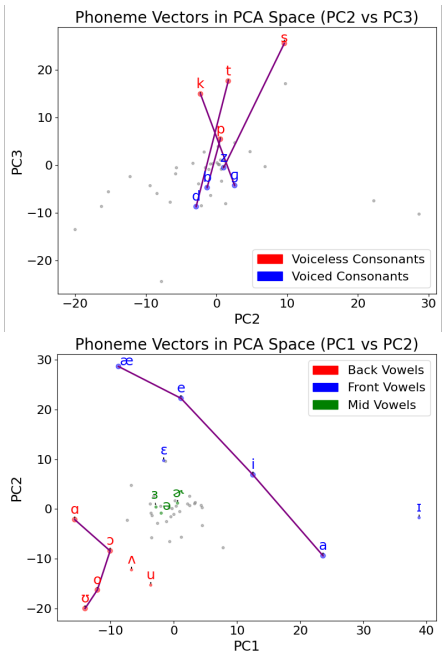

Figure 4: Phoneme vectors reduced to 2 dimensions using PCA trained on H13L12 result vectors. Phonetic patterns emerge among consonant voicedness (top) and vowel backness (bottom).

We visualize the result vectors of Head H13L12 for 5742 different target words on our template with principal component analysis (PCA) to understand how the model organizes phonetic information. We expect that if the model has an organized representation of phonetic information, we should see similar phonemes cluster together regardless of their source word.

For consonants, Principal Components 2 and 3 contain the least noisy phonetic information, in line with findings from Engels et al. (2025). Figure 4 shows that phoneme vectors corresponding to voiced consonants versus their voiceless equivalents follow a consistent linear pattern across PC3.

Vowels also follow consistent patterns in this space. These